

Project Overview

- **Goal:** Implement Reinforcement Q-learning for Mean Field Games in Finite Horizon.
- **Relevance:** Modeling agent decision making in order to model large population interactions (ex. investors/firms executing, investing, etc.)

Reinforcement Q-Learning

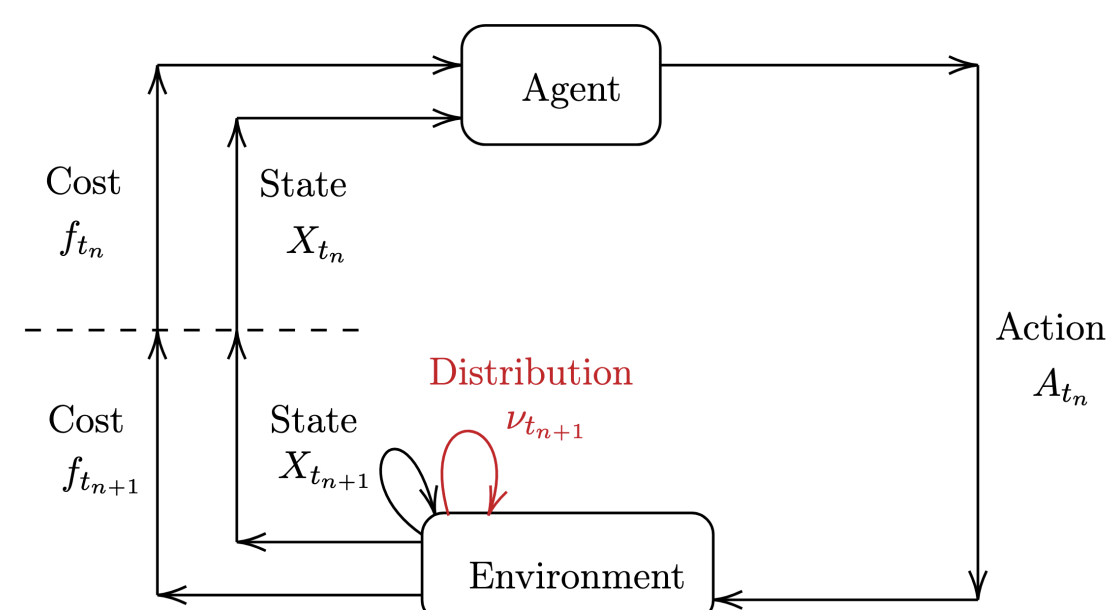
- **Reinforcement Learning:** Machine learning method in which an agent repeatedly interacts with an environment and learns based off of reward/cost signals from said environment.
- **Q-Learning:** Define a value function for each possible state-action pair as discounted expected future rewards. See Bellman equation:

$$Q_t(X_t, A_t) = f_t + \gamma \cdot \arg \min_{a'} Q_{t+1}(X_{t+1}, a')$$

Mean Field Game

- **Concept:** We model a population of identical competitive agents as a distribution (the mean field) that aim to find the optimal policy to maximize a reward/minimize a cost.
- **Trajectory method:** We model a singular agent, using their own trajectory to update the mean field distribution.

Figure 1: Reinforcement Learning on MFG



- **Note:** For MFG $\rho_\mu \ll \rho_Q$ while the converse is true for MFC.
- **Goal:** We seek an optimal policy for agents in a population of identical agents, a Nash Equilibrium
- We update the Q-function faster than the mean field, modeling the agent's control over its policy in reaction to the mean field.

Wealth Accumulation Problem

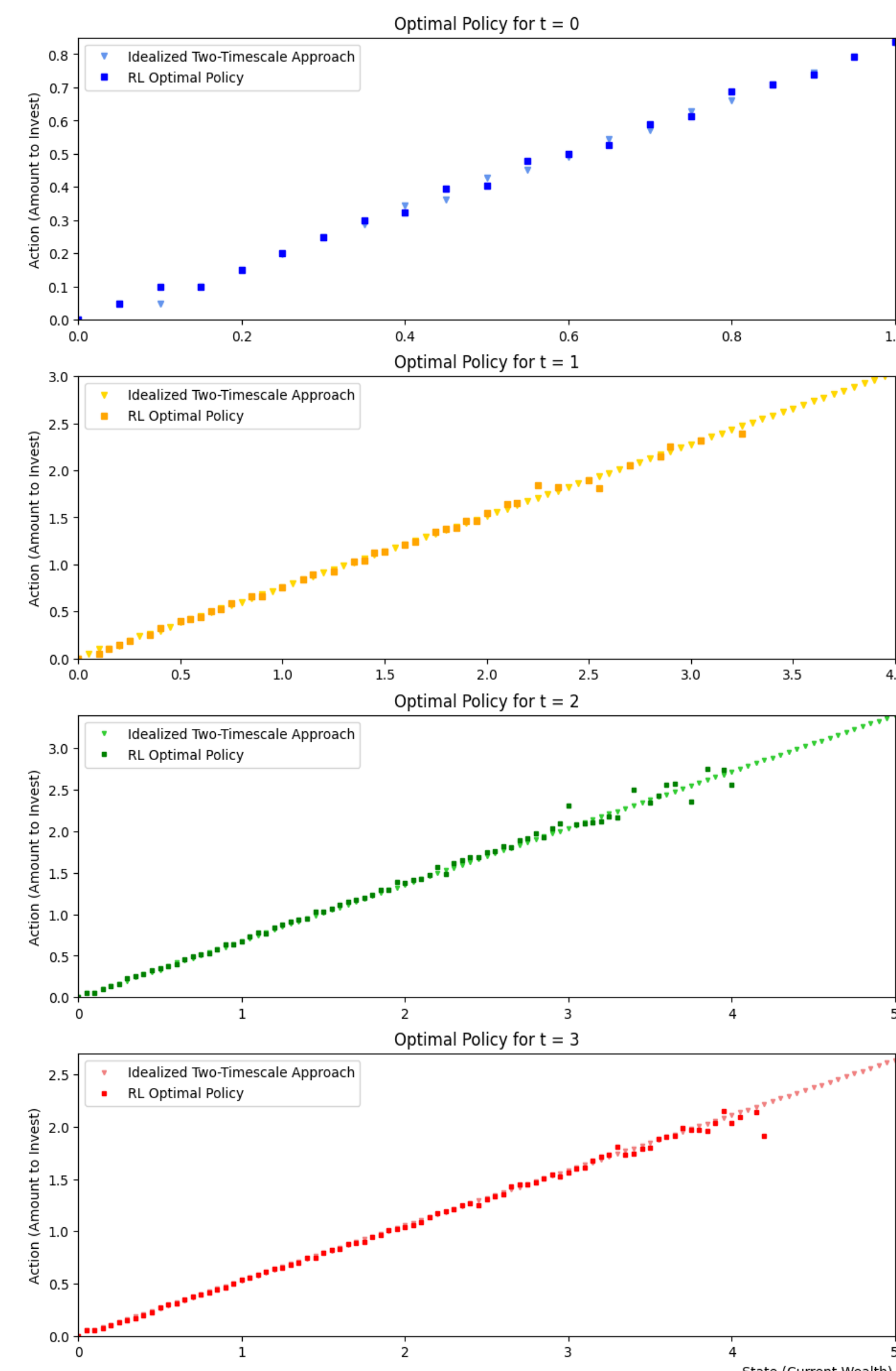
- **Context:** At discrete time steps, agents choose how much of their wealth to invest. As the mean field invests more wealth, investments have lower return.
- Uninvested wealth is consumed and gives a reward; invested wealth produces wealth for the next time step.

$$f(X_t, A_t) = \frac{1}{\gamma} (X_t - A_t)^\gamma$$

$$X_{t+1} = G \left(\int a d\theta_t(a), W_t \right) \cdot A_t$$

- We note that in the theoretical optimal policy, state and action are linearly related^[1].

Figure 2: Optimal Policy learned by RL



Price Impact Trader Problem

- **Context:** Learn the optimal execution in high-frequency trading when a large group of investors aim to buy or sell before a fixed horizon ($T = 1$).

- **System Dynamics:**

$$dX_t = \alpha_t dt + \sigma dW_t$$

$$dS_t = \gamma \left(\int_R a d\theta_t(a) \right) dt + \sigma_0 dW_t^0$$

$$dV_t = dK_t + X_t dS_t + S_t dX_t$$

(K_t denotes cash holdings)

- **Running Cost:**

$$f(t, x, \theta, \alpha) = c_\alpha(\alpha) + c_X(x) - \gamma x \int_R a d\theta_t(a)$$

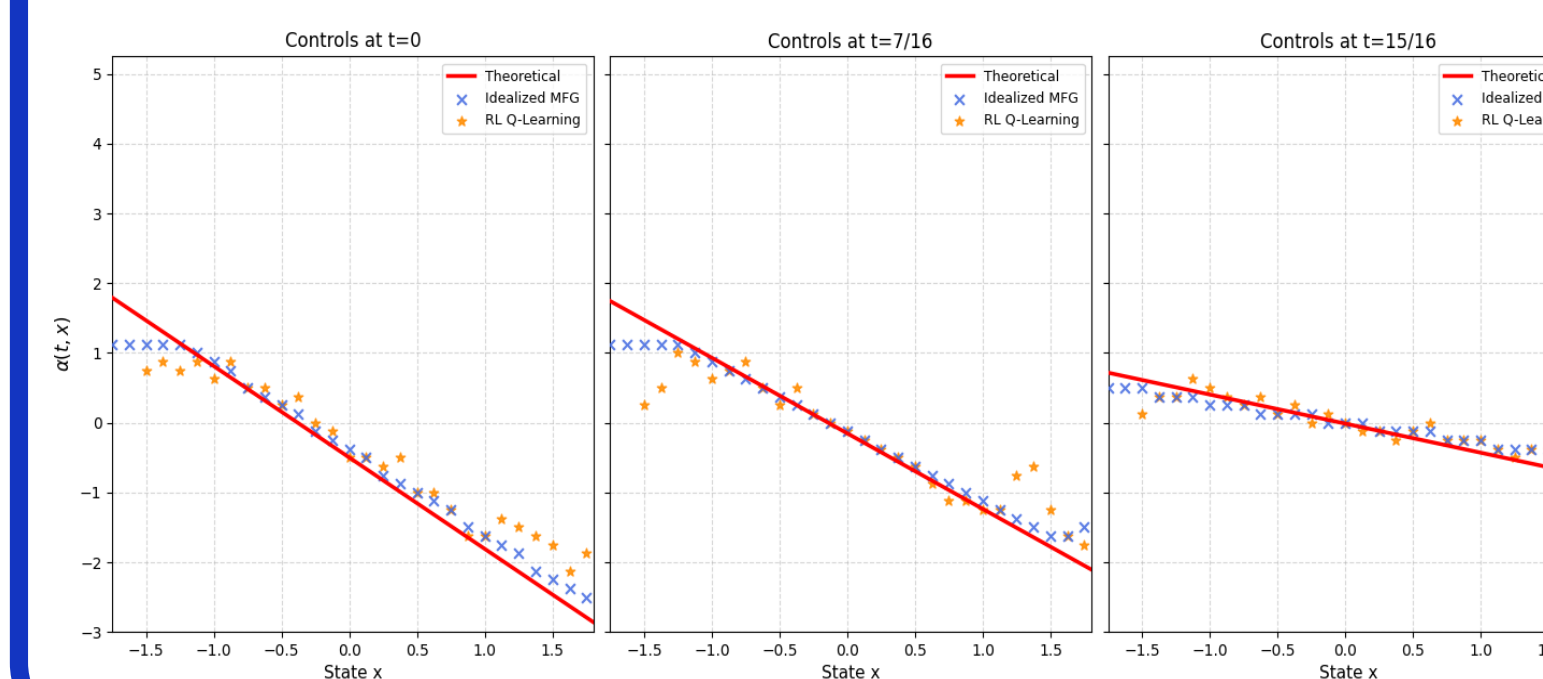
$$J(\alpha) = E \left[\int_0^T f(t, X_t, \theta_t, \alpha_t) dt + g(X_T) \right]$$

- **Solution:** The MFG equilibrium is defined as a pair that minimizes our cost function. Take a look at [1] for more information on how to solve yielding the optimal control:

$$-\hat{\alpha}(t, x) = -\frac{1}{c_\alpha} (\eta_t x + (\bar{\eta}_t - \eta_t) \bar{x}_t)$$

- The optimal strategy is **linear and decreasing in inventory**.
- The mean-field interaction enters only through the **average trading rate**. The congestion effect from the mean field depends only on the mean action.

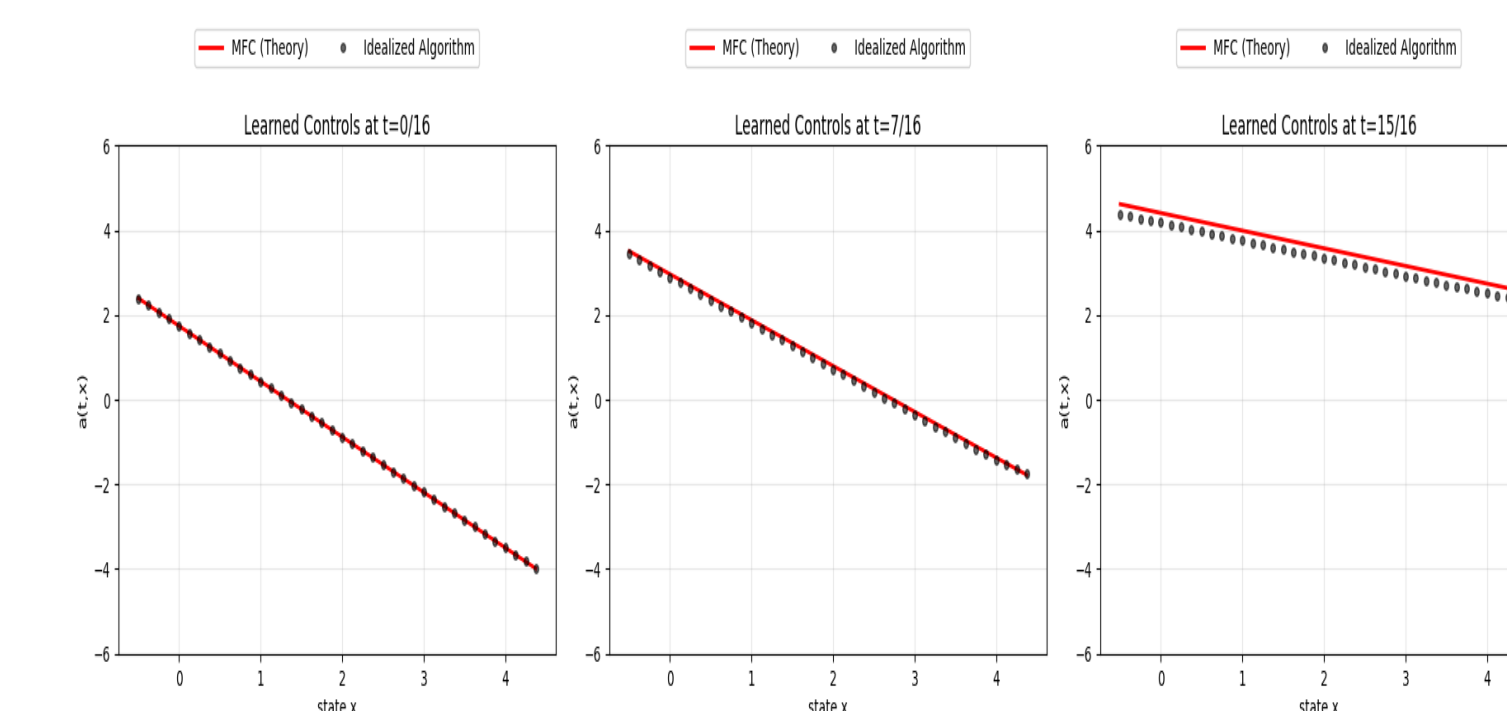
Figure 3: Optimal Controls



Idealized Two-Timescale

- **Idealized Approach:** Treat system as deterministic, update mean field distribution and Q-function by calculating expectations^[2].
- **MFC Idealized Algorithm:**
- We applied this to the MFC Price Impact Trader Problem and the results are shown below

Figure 4: Idealized Optimal Controls



Conclusion

- Extending this work would be implementing RL and Idealized Two-Timescale Approach in Finite Horizon Mean Field Control.
- MFC problems require higher computational complexity to solve due to keeping track of each state-action trajectory pair

References

1. Angiuli, A., Fouque, J.P, Laurière, M. (2021). Reinforcement Learning for Mean Field Games, with Applications to Economics. ArXiv, abs/2106.13755.
2. Angiuli, A., Fouque, J.P, Lauriere, M., Zhang, M. (2023). Convergence of multi-scale reinforcement Q-learning algorithms for mean field game and control problems. arXiv:2312.06659
3. Angiuli, A., Fouque, J. P., Laurière, M., Zhang, M. (2026). Analysis of multiscale reinforcement Q-learning algorithms for mean field control games. Applied Mathematics Optimization, 93(2), 27.